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Section:- 4A

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Assignment:- 1

QUESTION 1:-

Perform the multiplication $(762)_8 \times (45)_8$

Sol:-

$$\begin{array}{r} 762 \\ 45 \\ \hline 4672 \\ 3710x \\ \hline 43772 \text{ Ans} \end{array}$$

$$5 \times 2 = 10 \Rightarrow 8 + 2 = 12$$

$$5 \times 6 + 1 = 31 \Rightarrow 24 + 7 = 37$$

$$5 \times 7 + 3 = 38 \Rightarrow 32 + 6 = 46$$

$$4 \times 2 = 8 \Rightarrow 8 + 0 = 10$$

$$4 \times 6 + 1 = 25 \Rightarrow 24 + 1 = 31$$

$$4 \times 7 + 3 = 31 \Rightarrow 24 + 7 = 37$$

QUESTION 2:-

Perform the addition $(59F)_{16} + (E46)_{16}$

Sol:-

$$\begin{array}{r} 59F \\ E46 \\ \hline 13E5 \text{ Ans} \end{array}$$

$$15 + 6 = 21 \Rightarrow 16 + 5$$

$$9 + 4 = 14 \Rightarrow E$$

$$5 + 14 = 19 \Rightarrow 16 + 3$$

QUESTION 3:-

Convert the following numbers and complete the table -

Sol:-

The first row of the table
 $369.3125_{10} \rightarrow ?_2$

$$369/2 = 184 \quad 1$$

$$184/2 = 92 \quad 0$$

$$92/2 = 46 \quad 0$$

$$46/2 = 23 \quad 0$$

$$23/2 = 11 \quad 1$$

$$11/2 = 5 \quad 1$$

$$5/2 = 2 \quad 1$$

$$2/2 = 1 \quad 0$$

$$1/2 = 0 \quad 1$$

$$(369)_{10} \rightarrow (101110001)_2$$

Now after the decimal point

$$0.3125 \times 2 = 0.625 \quad 0$$

$$0.625 \times 2 = 1.250 \quad 1$$

$$0.25 \times 2 = 0.5 \quad 0$$

$$0.5 \times 2 = 1$$

$$(0.3125)_{10} \rightarrow (0.0101)_2$$

So

$$(369.3125)_{10} \rightarrow (101110001.0101)_2$$

Now to octal

$$101110001.0101_2 \rightarrow ?_8$$

$$\begin{array}{cccccc} \underbrace{101}_5 & \underbrace{110}_6 & \underbrace{001}_1 & \underbrace{010}_2 & \underbrace{100}_4 & \\ \hline & & & & & \end{array}$$

$$(101110001.0101)_2 \rightarrow (561.24)_8$$

Now to hexadecimal

$$101110001.0101_2 \rightarrow ?_{16}$$

$$\begin{array}{cccc} \underbrace{0001}_1 & \underbrace{0111}_7 & \underbrace{0001}_1 & \underbrace{0101}_5 \\ \hline & & & \end{array}$$

$$(101110001.0101)_2 \rightarrow (171.5)_{16}$$

Second row of the table

$$(1011101.101)_2 \rightarrow ?_{16}$$

$$1011101.101 = 2^7 + 2^5 + 2^4 + 2^3 + 2^2 + 2^0 + 2^{-1} + 2^{-3}$$

$$= 189.625$$

$$(1011101.101)_2 \rightarrow (189.625)_{16}$$

$$(1011101.101)_2 \rightarrow ?_8$$

$$\begin{array}{cccc} 010 & 111 & 101 & .101 \\ \hline 2 & 7 & 5 & 5 \end{array}$$

$$(1011101.101)_2 \rightarrow (275.5)_8$$

$$(1011101.101)_2 \rightarrow ?_{16}$$

$$\begin{array}{ccc} 1011 & 1101 & .101 \\ \hline B & D & A \end{array}$$

$$(1011101.101)_2 \rightarrow (BD.A)_{16}$$

Third row of the table

$$(326.5)_8 \rightarrow ?_2$$

$$3 \rightarrow 011 \quad 2 \rightarrow 010 \quad 6 \rightarrow 110 \quad 5 \rightarrow 101$$

$$(326.5)_8 \rightarrow (011010110.101)_2$$

$$(11010110.101)_2 \rightarrow ?_{16}$$

$$1101 \ 0110.101 \rightarrow D6.A$$

$$(11010110.101)_2 \rightarrow (D6.A)_{16}$$

$$(326.5)_8 \rightarrow ?_{10}$$

$$326.5 = 3 \cdot 8^2 + 2 \cdot 8^1 + 6 \cdot 8^0 + 5 \cdot 8^{-1}$$

$$= 214.625$$

Fourth row of the table

$$(F3C7.A)_{16} \rightarrow ?_{10}$$

$$\begin{aligned} F3C7.A &= F \cdot 16^3 + 3 \cdot 16^2 + C \cdot 16^1 + 7 \cdot 16^0 + A \cdot 16^{-1} \\ &= 15 \cdot 16^3 + 3 \cdot 16^2 + 12 \cdot 16^1 + 7 \cdot 16^0 + 10 \cdot 16^{-1} \\ &= 62407.625 \end{aligned}$$

$$(F3C7.A)_{16} \rightarrow (62407.625)_{10}$$

$$(F3C7.A)_{16} \rightarrow ?_2$$

$$F \rightarrow 1111 \quad 3 \rightarrow 0011 \quad C \rightarrow 1100 \quad 7 \rightarrow 0111 \quad A \rightarrow 1010$$

$$(F3C7.A)_{16} \rightarrow (11110011110001111010)_2$$

$$(11110011110001111010)_2 \rightarrow ?_8$$

$$\begin{array}{ccccccc} 001 & 111 & 001 & 111 & 000 & 111 & 101 \\ \hline 1 & 7 & 1 & 7 & 0 & 7 & 5 \end{array}$$

$$(11110011110001111010)_2 \rightarrow (171707.5)_8$$

Decimal	Binary	Octal	Hexadecimal
369.3125	10110001.0101	561.24	171.5
189.625	1011101.101	275.5	BD.A
214.625	1101010.101	326.5	D6.A
62407.625	11110011110001111010	171707.5	F3C7.A

QUESTION 4:-

Convert the following decimal numbers to indicated bases.

a) 7562.45 to octal

Sol:-

$$7562.45 \rightarrow ?_8$$

$$7562/8 = 945 \quad 2$$

$$945/8 = 118 \quad 1$$

$$118/8 = 14 \quad 6$$

$$14/8 = 1 \quad 6$$

$$1/8 = 0 \quad 1$$

$$(7562)_{10} \rightarrow (16612)_8$$

$$0.45_{10} \rightarrow ?_8$$

~~0.45~~

$$0.45 \times 8 = 3.60 \quad 3$$

$$0.60 \times 8 = 4.8 \quad 4$$

$$0.8 \times 8 = 6.4 \quad 6$$

$$0.4 \times 8 = 3.2 \quad 3$$

$$0.2 \times 8 = 1.6 \quad 1$$

$$0.6 \times 8 = 4.8 \quad 4$$

$$0.8 \times 8 = 6.4 \quad 6$$

$$\vdots \quad \vdots \quad \vdots$$

$$(0.45)_{10} \rightarrow (0.3461346\ldots)_8$$

$$(7562.45)_{10} \rightarrow (16612.3463146\ldots)_8$$

b) 1938.257 to hexadecimal

sol:-

$$(1938.257)_{10} \rightarrow ?_{16}$$

$$1938/16 = 121 \quad 2$$

$$121/16 = 7 \quad 9$$

$$7/16 = 0 \quad 7$$

$$(1938)_{10} = (792)_{16}$$

Now after the decimal part

$$(0.257)_{10} \rightarrow ?_{16}$$

$$0.257 \times 16 = 4.112 \quad 4$$

$$0.112 \times 16 = 1.792 \quad 1$$

$$0.792 \times 16 = 12.672 \quad 12 \text{ or } C$$

$$0.672 \times 16 = 10.752 \quad A$$

$$0.752 \times 16 = 12.032 \quad C$$

$$0.032 \times 16 = 0.512 \quad 0$$

$$0.512 \times 16 = 8.192 \quad 8$$

....

Here we can stop

$$(0.257)_{10} \rightarrow (0.41CAC08\dots)_{16}$$

the final answer is

$$(1938.257)_{10} \rightarrow (792.41CAC08\dots)_{16}$$

c) 175.175 to binary

$$175/2 = 87 \quad 1$$

$$87/2 = 43 \quad 1$$

$$43/2 = 21 \quad 1$$

$$21/2 = 10 \quad 1$$

$$10/2 = 5 \quad 0$$

$$5/2 = 2 \quad 1$$

$$2/2 = 1 \quad 0$$

$$1/2 = 0 \quad 1$$

$$(175)_{10} = (10101111)_2$$

Now the decimal part

$$0.175 \times 2 = 0.35 \quad 0$$

$$0.35 \times 2 = 0.7 \quad 0$$

$$0.7 \times 2 = 1.4 \quad 1$$

$$0.4 \times 2 = 0.8 \quad 0$$

$$0.8 \times 2 = 1.6 \quad 1$$

$$0.6 \times 2 = 1.2 \quad 1$$

$$0.2 \times 2 = 0.4 \quad 0$$

$$(0.175)_{10} \rightarrow (10101111.00101100\dots)_2$$

QUESTION 5:-

a) Write the digits for a base 20 systems.

Sol:-

0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F
16	G
17	H
18	I
19	J

b) Convert $(2007)_{10}$ to base 20

Sol:-

20	2007	
	100	7
	5	0

$$(507)_{20}$$

c) Convert $(BCI.G)_{20}$ to decimal

Sol:-

$$\begin{aligned} &= B \times 20^2 + C \times 20^1 + I \times 20^0 + G \times 20^{-1} \\ &= 11 \times 20^2 + 12 \times 20^1 + 18 \times 20^0 + 16 \times 20^{-1} \\ &= 4658.8 \end{aligned}$$

$$(BCI.G)_{20} \rightarrow (4658.8)_{10}$$

QUESTION 6:-

Represent the decimal numbers in BCD.

a) 694

Sol:-

As we have

$$4 = 0100, \quad 9 = 1001, \quad 6 = 0110$$

so

$$694 = (011010010100)_{BCD}$$

b) 835

Sol:-

As we have

$$8 = 1000, 3 = 0011, 5 = 0101$$

So

$$835 = (100000110101)_{BCD}$$

Now their sum

$$011010010100$$

$$100000110101$$

$$111011001001$$

$$+ 0110$$

$$11100010$$

$$0110$$

$$10101$$

$$0001 \quad 0101 \quad 0010 \quad 1001$$

$$\downarrow$$

$$\downarrow$$

$$\downarrow$$

$$\downarrow$$

$$1$$

$$5$$

$$2$$

$$9$$

By summing the answer is 1529

QUESTION 7:-

Decode the following ASCII codes:-

Sol:-

Decimal

ASCII code

$$1000111$$

$$71$$

$$G$$

$$1101111$$

$$111$$

$$0$$

$$0100000$$

$$32$$

$$\text{SPACE}$$

$$1000010$$

$$66$$

$$B$$

$$1100001$$

$$97$$

$$a$$

$$1100100$$

$$100$$

$$d$$

1100 111	103	g
1100 101	101	e
11100 10	114	n
11100 11	115	s
010000 1	33	!

QUESTION 8:-

Write your full name in ASCII using 8-bit code

a) with the left most bit always zero

sol:-

S	01010011
H	01001000
E	01000101
Z	01011010
A	01000001
space	00100000
N	01001110
A	01000001
V	01010110
E	01000101
E	01000101
D	01000100

b) with the leftmost bit selected to produce even parity

sol:-

S	11010011
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H	11001000
E	11000101
Z	11011010
A	11000001
SPACE	10100000
N	11001110
A	11000001
V	11010110
E	11000101
E	11000101
D	11000100